

DRISHTI

Final 10-minute, six-member speech

Camera-primary navigation for GPS-denied outdoor UGVs

The gist

DRISHTI is a proposed software navigation module for an outdoor ground robot. We will adapt selected components from LARIAD's Offroad-Nav baseline: camera images provide depth and terrain cues, while a visual-inertial estimator tracks motion and supplies sparse metric references for depth alignment. A rolling local map records estimated ground, obstacles, support, invalid regions, and observation age. A registered route and destination guide a predictive planner; a supervisor checks the vehicle footprint, data validity, and stopping space before allowing motor commands. The intended responses are to proceed, slow down, reroute, or stop. The first demonstration uses a previously mapped daylight route without operational GPS. Auxiliary-sensor permission, deployment performance, and physical reliability must still be validated. CPU-only operation is an optimization objective, not a demonstrated result.

Requirement basis: [1, 12]. Research baseline: [2-7]. The integration, supervision and evaluation are proposed DRISHTI work.

Speaking order and timing

Time	Speaker	Focus	Words
0:00-1:30	Juhil Modi	Problem and proposal	191
1:30-3:10	Saswata Das	Offroad-Nav and contribution	210
3:10-4:50	Harshit Tiwari	Perception and map validity	218
4:50-6:30	Rohan Jangam	Localization and control	214
6:30-8:10	Ritesh Prajapati	Implementation and evaluation	213
8:10-9:40	Sanskriti Kumari	Deliverable and closing	190

READ ALOUD: PAGES 3-8 ONLY

1,236 spoken words, including handoffs. Rehearse at about 130-140 words per minute with deliberate pauses. Target completion at 9:40; retain 20 seconds of buffer. Actual timing must be checked in rehearsal. The other pages are preparation notes, not extra speech.

This is a speech and rehearsal packet, not a six-slide deck.

Yes: Offroad-Nav is our baseline

The commitment is to use and credit selected Offroad-Nav research and code, not merely cite its name. The planned starting point is its monocular depth-rescaling and terrain-mapping path. Reproduction, adaptation, and integration are future engineering milestones; do not imply that the complete stack is already running on your robot. [2-4]

Area	Offroad-Nav reference	DRISHTI decision
Depth and mapping	Depth Anything V2, sparse-reference rescaling, boundary filtering, temporal smoothing and elevation mapping. [2, 3]	Audit and adapt the relevant components; attach validity checks, support status and observation age.
Localization	The paper reports GNSS, IMU and SLAM fusion; sparse rescaling references use VINS-Mono. [3]	Use ORB-SLAM3 monocular-inertial as the initial GPS-free candidate. This replacement requires its own tests.
Navigation runtime	Public stack targets ROS 1 Noetic, move_base/TEB and GPU elevation mapping. [4]	Reproduce a pinned upstream baseline separately. Target selective ROS 2 integration with Nav2 MPPI, not an overnight whole-repository port.
Proposed additions	Traversability and negative obstacles are identified as improvement directions. [3]	Add PIDNet-S terrain cues, explicit unknown/support states, vehicle-specific motion checks and controlled ablations.
Deployment proof	A published system is not a benchmark of our hardware.	Measure the full adapted pipeline first. CPU-only deployment, FPS, memory and cost remain unverified.

The configuration all six speakers must use

One RGB camera + synchronized IMU; encoders for control and consistency checks, subject to sensor approval. Depth Anything V2 Small is the reference depth model; PIDNet-S is the terrain-semantic starting point. ORB-SLAM3 provides the initial localization candidate. The rolling map, route registration, Nav2 integration and command supervisor are our proposed engineering work. [5, 7, 8, 10]

EXCLUDED FROM THE COMMITTED PROTOTYPE

No mandatory RL policy, general neural world simulator, joint nine-model architecture, operational GNSS/LiDAR/stereo input, universal terrain guarantee, or unmeasured CPU-only promise. Compression and learned dynamics can be evaluated later.

The original public repository currently marks the physical-robot branch as unavailable. It is a research/code starting point, not a turnkey hardware handover. [4]

Juhil Modi

The problem and our proposed solution

Good morning. We are Team UniMinds, presenting DRISHTI for Bharat Electronics Limited's problem statement SIH26126: vision-based outdoor navigation for an unmanned ground vehicle.

Recognising a rock is not enough to navigate around it. The robot must estimate where the rock is, understand the ground, track its own movement, and choose a motion it can execute.

Our proposed software connects those tasks to move from Point A to Point B without operational GPS. It combines camera-based perception, visual-inertial localization, a compact terrain map, and predictive planning. The output is a route and supervised motion commands for an existing vehicle, not a new vehicle design.

We propose one calibrated RGB camera, supported by a synchronized IMU and wheel encoders. Permission for auxiliary sensors remains an organizer clarification. We are calling this camera-primary, not camera-only.

Our first physical demonstration will target a controlled daylight route on firm terrain, at low speed, with measured obstacles and protected drop-off tests. This establishes a bounded test before more difficult environments.

Today we are presenting the architecture and validation plan, not claiming completed field validation. Saswata will explain our research foundation and the work we propose to contribute.

END OF SPOKEN SECTION

Delivery cue: Lead with the navigation task, not a list of models.

Source notes, not spoken: [1], [12]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Saswata Das

Offroad-Nav: our baseline, not our invention

We will build on LARIAD's Offroad-Nav as our research and code baseline. We plan to audit and adapt selected depth-rescaling and terrain-mapping components, with attribution, rather than recreate the entire stack. Our repository will distinguish unchanged upstream code, modified components, and new DRISHTI modules, so the engineering contribution remains traceable.

Its monocular pipeline combines Depth Anything V2, sparse geometric depth references, elevation mapping, and conventional planning. This gives us a concrete starting point to reproduce and evaluate.

However, its reported localization includes GNSS, and the current public stack targets ROS 1. Those results are not proof of our GPS-free ROS 2 configuration. Removing satellite input requires a functioning replacement estimator and a registered destination, not simply deleting a sensor topic.

Our proposed contribution is to connect terrain-map validity with vehicle-specific motion decisions. The map will distinguish adequately observed ground from obstacles, stale observations, and unknown regions. The planner will consider the vehicle's footprint, clearance, and measured motion limits.

The research question is whether these checks reduce hazardous decisions and interventions without preventing useful route completion. We will compare them against the same navigation baseline.

A general neural world simulator and reinforcement-learning policy are not dependencies for this first deliverable. Harshit will explain how we turn images into usable terrain estimates.

END OF SPOKEN SECTION

Delivery cue: Say explicitly that Offroad-Nav is the selected starting baseline.

Source notes, not spoken: [2], [3], [4]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Harshit Tiwari

Depth, terrain and the quality of the evidence

Our reference depth model is Depth Anything V2 Small. Its standard relative-depth output is not a direct measurement in metres.

After valid visual-inertial initialization, we will use reliable sparse metric landmarks to align the dense prediction. The published method fits scale and shift in inverse depth. We will check reference coverage, residual errors, timestamps, and parameter stability before accepting the result.

This process does not make every pixel accurate. We will reject inconsistent estimates and restrict geometric decisions to the range supported by our tests.

Accepted depth is projected into three dimensions. A local ground estimate then helps distinguish obstacle elevation from camera distance. Unreliable boundary pixels remain unknown; filtering must not manufacture free space. For a ditch, missing geometry is not proof of a drop, but it is not proof of supporting ground either.

A separate semantic branch, starting with PIDNet-S, will use off-road RGB annotations from GOOSE and local recordings. It supplies terrain cues, not proof that grass can support this vehicle or that a ditch is absent.

The rolling map combines geometry, terrain cues, support status, observation age, and validity indicators. These indicators are not calibrated collision probabilities. Both branches share the same camera, so they are complementary rather than independent safety channels.

Rohan will explain how the map and localization guide the robot's motion.

END OF SPOKEN SECTION

Delivery cue: Distinguish relative depth, metric alignment and ground-relative height.

Source notes, not spoken: [5], [6], [7], [8], [9]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Rohan Jangam

Localization, destination and predictive control

Our initial localization candidate is ORB-SLAM3 in monocular-inertial mode. Camera calibration, camera-to-IMU alignment, synchronization, and initialization are explicit engineering tasks.

During a supervised mapping pass, we will register visual landmarks, the route, and Point B in a common coordinate frame. Autonomous execution will then use that map without GPS. This is GPS-free teach-and-repeat with local replanning, not unrestricted map-free exploration.

If tracking fails or the destination loses its valid relationship to the current map, we suspend goal pursuit. Avoiding nearby obstacles does not repair a lost global destination.

Global route planning and local control remain separate. For local control, Nav2's MPPI controller predicts candidate vehicle motions and evaluates them against the route and obstacle costs. It does not require reinforcement learning. The selected motion model and constraints must match the actual chassis.

Our supervisor will additionally check the swept footprint, sensing freshness, terrain support, and stopping space before accepting a command. A newly detected obstacle should trigger slowing or stopping, followed by replanning when an observed detour is available.

Initially, we disable automatic reversing into unseen space. Speed limits depend on measured latency and braking, and a motor watchdog expires stale commands. Physical tests retain an emergency stop; these protections are not a collision-free guarantee.

Ritesh will describe how we evaluate the complete loop.

END OF SPOKEN SECTION

Delivery cue: Explain how Point B stays registered and what happens when tracking fails.

Source notes, not spoken: [7], [10]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Ritesh Prajapati

Implementation, experiments and resource reality

We will build in stages: recorded-data checks, closed-loop simulation, and then low-speed physical trials. Reproduction of Offroad-Nav and validation of our modified configuration are separate milestones.

In the full GPS-free simulation test, ground-truth pose and depth will go only to the evaluator, not the navigation pipeline. Any diagnostic run using privileged inputs will be labelled separately.

Our ROS 2 implementation will use C++ for timing-sensitive modules and Python for training and evaluation. Sensor wrappers, coordinate transforms, map interfaces, and chassis control require integration; this is not a plug-and-play claim.

We will report completion, collisions, interventions, destination error, tracking failures, unnecessary stops, and sensing-to-command latency. We will predefine arrival tolerance, timeout, collision criteria, and intervention rules, then report every trial rather than selected demonstrations. Tests will include newly introduced soft obstacles, protected drop-offs, frozen images, and localization loss.

Held-out recordings and routes will separate tuning from evaluation. Comparisons will keep sensing, localization, and the planner fixed while adding terrain cues and our validity policy. Physical trials retain identical watchdog and emergency-stop protections.

CPU-only deployment is an optimization objective, not an established result. We will first measure the baseline, then test compression against navigation outcomes. Offline video accuracy will not be presented as physical autonomy.

Sanskriti will conclude with the deliverable and its intended value.

END OF SPOKEN SECTION

Delivery cue: A test plan is stronger than unsupported accuracy or FPS claims.

Source notes, not spoken: [4], [5], [7], [10]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Sanskriti Kumari

Deliverable, intended impact and closing

DRISHTI's deliverable will be a navigation module with calibration instructions, launch configuration, evaluation logs, and a documented operating envelope.

An operator view will show the route, terrain state, localization health, and the reason for a slowdown or stop. Navigation is intended to run onboard without a cloud dependency. Reducing compute comes after demonstrating useful closed-loop behaviour.

Our immediate users are teams integrating outdoor UGVs. Inspection, agricultural logistics, and delivery are potential applications after validation for their conditions, not environments we claim to have solved already.

We will credit Offroad-Nav and the other reused components, and review their individual licences before distribution. Our proposed work is the GPS-free integration, vehicle-aware terrain decisions, validity supervision, and controlled evaluation, not ownership of the underlying published algorithms.

The central trade-off is measurable: a robot that stops everywhere does not complete the task, while a robot that continues on unreliable evidence can make dangerous decisions. We will report both completion and failures.

DRISHTI aims to answer two questions together: where should the vehicle go, and do its current observations support the next manoeuvre?

Thank you. We welcome questions about the implementation, assumptions, and evaluation plan.

END OF SPOKEN SECTION

Delivery cue: Close on useful autonomy and transparent evidence, not universal reliability.

Source notes, not spoken: [1], [2], [3]. Published component capabilities are distinguished from proposed DRISHTI integration and testing.

Questions the whole team must answer

Are we actually using Offroad-Nav?

Yes, as the selected research/code baseline for planned adaptation. Begin with the monocular rescaling implementation and the depth-to-elevation-map reference path; pin the upstream revision and retain attribution. The final ROS 2 system will not be the unchanged upstream stack. Say 'we will adapt' until your repository and tests establish what has actually been integrated. [2-4]

What is our contribution, rather than borrowed work?

GPS-free integration, robot-conditioned terrain decisions, explicit handling of invalid or stale observations, command supervision, and controlled evaluation. The hypothesis is that these additions improve the completion-versus-risk trade-off at a measured compute cost. Do not claim global algorithmic novelty or superiority before a comparison.

How does one camera give metric depth?

The proposed camera-plus-IMU estimator supplies sparse metric references only after adequate calibration, initialization and motion excitation. Fit $1/D = a \cdot q + b$, where q is the relative inverse-depth prediction and D is optical-axis depth. Check residuals and positive, usable denominators. Scale alignment cannot guarantee correct shape at every pixel. Pure scale-ambiguous monocular landmarks are not a metric reference. [6, 7]

Are the IMU and encoders definitely allowed?

The statement specifies camera feeds as the primary sensor; it does not explicitly settle every auxiliary-sensor restriction. Obtain organizer confirmation. The camera is the environmental sensor, the IMU supports motion estimation, and encoders support control and consistency checks. Do not call this camera-only, and do not promise an equivalent camera-only fallback without testing it. [1]

What happens when localization fails?

Suspend goal pursuit, stop through the supervised command path, and resume only after a valid map association is restored. A new disconnected SLAM map is not automatically linked to Point B. Local obstacle avoidance cannot repair an invalid global goal. Sparse SLAM landmarks and the terrain costmap are different representations that must share a consistent frame. [7; proposed policy]

Scope, evidence and practical delivery

Does a prior mapping pass make this non-autonomous?

The proposed execution phase localizes and controls itself on a registered route, including local obstacle handling. That is autonomous teach-and-repeat, not map-free exploration. The route and goal are registered in a visual-map frame; a sparse SLAM map alone is not a Nav2 traversability grid. This first stage does not establish general navigation in unseen environments.

Can we detect every ditch or predict moving people?

No. Test visible drop-offs and regions where ground support is not established; concealed hazards remain a limitation. MPPI predicts candidate vehicle motions, not automatically pedestrian trajectories. Initially handle newly observed obstacles with low-speed stop/replan behaviour; do not promise unrestricted dynamic-scene prediction. [10; proposed scope]

Is CPU-only real-time operation already proven?

No. Establish a working reference and measure the complete sensing-to-command chain on named hardware. Then evaluate a smaller model, resolution changes and quantization against hazard and navigation outcomes. Quantization can affect accuracy and does not guarantee a speedup on every CPU. Never present controller frequency as fresh camera-update frequency. [5, 11]

What proves the system works?

Define arrival tolerance, timeout, collision and intervention rules before trials. Report all held-out runs, localization failures, unnecessary stops and latency. Separate offline perception, diagnostic simulation, sensor-only closed-loop simulation, and physical tests. Keep emergency stops and watchdogs identical across physical comparisons; test unsafe-policy ablations in simulation.

Rehearsal rules

Use the same architecture and tense across all six voices. Mark every number, screenshot and clip as our measured result, published reference, or planned target. Do not use borrowed footage as your demonstration. A clip must replace speaking time, not be added to ten minutes. Keep the PDF available offline and rehearse the handoffs with a timer.

WHEN ASKED WHAT IS FINISHED

Answer with the actual repository, test logs and hardware evidence. Use Planned / Running / Measured consistently. This packet commits to a proposed design; it does not certify completed integration or field reliability.

Research and verification references

Checked 8 September 2026. Numbered notes on the speech pages refer to these sources. Sources support the requirements and component descriptions, not unperformed DRISHTI experiments.

[1] Smart India Hackathon 2026 problem statements

Official SIH portal; SIH26126, Bharat Electronics Limited, Software / Smart Automation. [Official problem listing](#)

[2] Offroad-Nav project page

LARIAD / ENSTA-AMIAD, 2026. Overview of monocular perception and off-road navigation. [Project website](#)

[3] An Open-Source LiDAR and Monocular Off-Road Autonomous Navigation Stack

Marsal et al., arXiv:2604.03096, 2026 preprint. See method, implementation and conclusion. [Paper](#)

[4] LARIAD / Offroad-Nav

Official repository. Current ROS target, dependency structure, branch status and licensing. [Code and README](#)

[5] Depth Anything V2

Yang et al., NeurIPS 2024; official repository and Small-model checkpoint documentation. [Model repository](#)

[6] A Simple yet Effective Test-Time Adaptation for Zero-Shot Monocular Metric Depth Estimation

Marsal et al., IROS 2025; arXiv:2412.14103. Sparse-reference inverse-depth alignment. [Rescaling paper](#)

[7] ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM

Campos et al., IEEE Transactions on Robotics, 2021. Official implementation and initialization discussion. [Paper](#) | [Code](#)

[8] PIDNet: A Real-Time Semantic Segmentation Network Inspired by PID Controllers

Xu et al., CVPR 2023. Official code; PIDNet-S is a candidate to adapt, not an off-road result. [Official repository](#)

[9] GOOSE dataset

Official project: RGB semantic annotations for off-road scenes, with additional modalities. [Dataset website](#)

[10] Nav2 Model Predictive Path Integral Controller

Official Jazzy documentation. Candidate motion rollout, motion models and configurable costs. [MPPI documentation](#)

[11] ONNX Runtime: Quantize ONNX Models

Official documentation. Quantization workflow, accuracy considerations and hardware-dependent performance. [Quantization documentation](#)

[12] Supplied problem statement and team details

sih2026.txt: problem statement, lines 1-24; six-member names, lines 147-155. The surrounding transcript contains earlier proposals, not proof of implementation.